

Predicting Used Toyota Corolla Prices

Model Development, Diagnostics, and the Limits of a Causal Reading

Final Project · Data Analysis with Statistical Software · Summer 2026

1. Executive Summary and Research Objectives

This project answers two questions that look alike and are not. **Parts A–C are predictive:** how accurately can the price of a used Corolla be predicted? The criterion is out-of-sample error in euros, and variables enter because they lower it — selected by cross-validation and regularisation, never by filtering p-values. **Part D is causal:** what is one vehicle characteristic actually worth? There the criterion is the absence of bias, and p-values return for a different job — testing whether the two groups of cars are comparable at all. Following Shmueli (2010) the two are kept apart throughout, because a model that wins the first question is not thereby entitled to answer the second.

The analysis set is **1,429 cars** (1,436 minus five Verso MPVs and two impossible weights), split 60/40 into 857 training and 572 holdout cars, with **45 base predictors**. Nineteen specifications were fitted under one protocol fixed in advance. The preferred model — the 45 predictors plus six forward-selected derived terms — records **RMSE 976.2 ± 15.3 €** in cross-validation and **1,088.6 € RMSE, 819.7 € MAE, 8.2% MAPE and $R^2 = 0.900$** on the 572 holdout cars, opened once, after the model had been named.

Two of the most useful results are negative and are reported as findings. **Regularisation buys nothing here:** on the final design both penalties collapse to the smallest value searched and finish 2–4 € behind unpenalised OLS, against a standard error of 15 €. And **constant variance fails in all nineteen runs without exception**. Four unrelated remedies were tried and none came close, which is itself the answer: a 20,000 € car is genuinely harder to price than a 5,000 € one. The response is HC3 robust standard errors — keep the predictions, repair the inference. Part D then takes *Automatic_airco* (digital climate control) as a treatment: the raw 8,622 € gap falls to somewhere between 1,210 € and 1,975 € depending on which defensible specification is used, and none of those numbers is the value of the option.

2. Part A — Data Preparation, Audit and the Design Matrix

2.1 Quality audit and the target population

The file is clean on the obvious checks: no missing cells, no duplicate rows, no duplicate identifier. Three structural problems surfaced instead. The identity $\text{Age} = 12(2004 - \text{Mfg_Year}) + (9 - \text{Mfg_Month})$ holds on **100% of rows** (auxiliary $R^2 = 1.000000$), so keeping all three columns would leave the matrix rank-deficient; the two date columns were dropped. Cylinders is 4 for every car and Parking_Assistant is on in 4 cars out of 1,436 — neither carries information to estimate.

The five Verso records were excluded as a **population definition, not as outlier removal**. They are seven-seat MPVs: 2.0 D4D, 1,320–1,480 kg against a Corolla median of 1,067 kg, priced up to 32,500 €. Five cars cannot support a depreciation curve of their own, and pooling them distorts the weight coefficient for the other 1,431. Re-including them later costs 2.5 € of cross-validated error — which confirms the exclusion was right for its stated reason rather than because it flattered the numbers.

2.2 Record-level corrections

Two data-entry errors were handled differently, according to whether the record contained its own correction. **CC = 16,000 on Id 81** is recoverable: the model string reads "1.6", so the value was set to 1,600. **Weight has no such anchor**, so it was audited within specification rather than against the whole fleet — a wagon is legitimately heavier than a hatchback. The rule was fixed before the result was inspected: among cars sharing HP, CC, Fuel_Type, Doors and Automatic, a weight more than 5 SD from the sub-group mean is a suspected error. Exactly two records break it — Id 223 at 1,615 kg ($z = 16.8$) and Id 964 at 1,480 kg ($z = 11.5$, exactly the Verso weight). The next-largest deviation is 4.5 SD, so the cut does not depend on where between 5 and 10 SD it sits. Both excluded; $n = 1,429$.

Finding, and it is the single most consequential decision in Part A. Putting those two records back costs **102 €** of cross-validated error and collapses the Weight coefficient from **17.36 € per kg to 2.97**. Two rows out of 859 nearly erase one of the strongest relationships in the data. **So what:** data quality is a larger lever here than model choice — the gap between the best and worst estimator in the whole study is smaller than the damage two typos do. The rule was declared in advance precisely so this result could not be reverse-engineered.

2.3 Feature engineering and encoding

The free-text Model column is unusable as a predictor but carries two recoverable tokens. *Trim* (TERRA / LUNA / SOL) was extracted with LINEA treated as a marketing prefix rather than a level, and *Body* through an alias table (HB, HATCHB, HATCHBACK all map to Hatchback). No record carries two trim tokens, so the extraction is unambiguous. The assumed ordering is not an assumption

— it is visible in three independent columns: air-conditioning fitted to 32% of TERRA cars, 83% of LUNA and 100% of SOL, with median prices of 9,000 €, 9,900 € and 13,500 €. Cars with no token were kept as an explicit *Unknown* level rather than discarded; they are the older G6/S/GTSi generation and sit consistently between LUNA and SOL on every column. Three further tokens were documented but not used, since D4D reproduces Fuel_Type and AUT3/AUT4 reproduce Automatic exactly. Radio and Radio_cassette agree on 1,426 of 1,429 rows ($r = 0.992$), so Radio was dropped; three colours with three or four cars each were merged into Other while White ($n = 31$) was kept — the criterion is estimability, not similarity of medians.

2.4 Split, verification and pre-registration

The data was split **60/40, stratified by price decile**, because price is skewed and a simple random split can leave the two sets with visibly different price profiles — the holdout would then measure the split rather than the model. The split was verified rather than assumed: Kolmogorov–Smirnov tests return **$D = 0.028$ ($p = 0.95$) on Price, 0.021 ($p = 1.00$) on Age and 0.024 ($p = 0.99$) on KM**, every trim and body level has at least 54 training cars, and the partition was hashed so no later cell could silently redraw it. The matrix holds 45 predictors (29 numeric or binary, 16 dummies), is full rank, and gives about 19 training observations per parameter. Two things are deliberately absent. Standardisation is not applied here — OLS is invariant to column scale, and standardising now would compute means over rows that later serve as validation folds, so the scaler lives inside the cross-validation pipeline and is refitted per fold. Nor are squares and interactions created: they are specification hypotheses belonging to Part C, and building them during data preparation would pre-empt the diagnosis.

Finally, the evaluation protocol was fixed before a single model was fitted: RMSE in euros as the primary metric; repeated 10-fold cross-validation (3 repeats, seed 2026) common to every candidate; paired Wilcoxon tests with Holm correction for any claim of superiority; a one-standard-error rule with a pre-declared simplicity order for breaking ties; and a holdout opened exactly once, after the preferred model has been named. The purpose is to remove the degrees of freedom that let an analyst choose, after the fact, the metric under which their favourite model happens to win.

3. Part B — Model Development and Regularisation

Three models were fitted on the same 45 predictors, the same 857 training rows and the *same folds*, so any difference between them is the method and not the split. Ridge penalises squared coefficients, shrinking them toward zero without reaching it, so all 45 survive and a collinear cluster is stabilised rather than resolved; LASSO penalises absolute coefficients, drives some to exactly zero, and so selects and estimates in one step. Because both penalties act on the coefficients themselves their bite depends on column scale — unstandardised, KM (tens of thousands) and Automatic (0/1) would be penalised on entirely different terms — so every penalised fit runs inside a pipeline that standardises using training-fold statistics only, and its coefficients read as euros per standard deviation.

3.1 Choosing the penalty

λ cannot be read off the data, so it was chosen like every other tuning decision here: a log-spaced grid, each value scored by cross-validated RMSE on the identical folds, lowest error winning. Both grids select $\lambda \approx 11.72$, and at that value LASSO retains 38 of the 45 predictors.

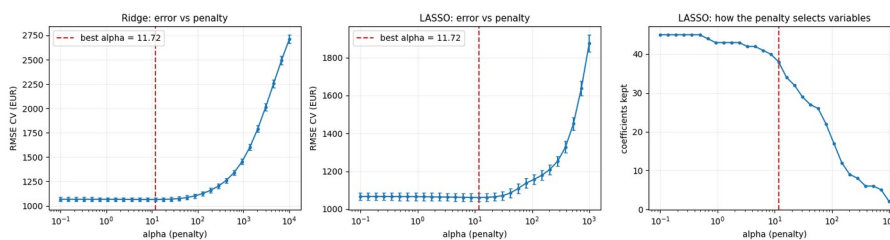


Figure 1. Cross-validated error against the penalty for Ridge (left) and LASSO (centre), and the number of surviving LASSO coefficients (right). Error is flat across the lower half of both grids; the penalty changes which variables survive long before it changes accuracy.

Finding. The error curves are flat until roughly $\lambda = 100$ and only then climb; by $\lambda = 10,000$ Ridge has degraded to about 2,700 €. **So what:** the choice of penalty is not delicate here — what is at stake is parsimony, not accuracy.

Model	k	RMSE CV (€)	SE	MAE CV (€)	Predictors kept
OLS	45	1,067.3	20.3	794.7	45
Ridge ($\lambda = 11.72$)	45	1,064.9	20.6	791.7	45
LASSO ($\lambda = 11.72$)	45	1,061.4	21.1	788.9	38

Table 1. The three Part B models on identical folds, 857 training cars.

3.2 What the comparison says

On prediction the three are indistinguishable — a 6 € spread against a standard error of about 20 €. With 857 rows against 45 predictors the estimates are already stable, leaving little variance for shrinkage to remove. **What LASSO buys is interpretation, not accuracy.** It zeroes 7 of the 45 coefficients, Fuel_Type_Diesel among them — the worst collinearity offender in the study, with VIF 23.5 and an auxiliary R^2 of 0.957 when regressed on HP, CC, Weight and Quarterly_Tax. Its behaviour across the three fits is the whole diagnosis in miniature: OLS puts it at +25 €, Ridge at -28 €, LASSO at exactly 0. A coefficient whose sign depends on which estimator you use is not measuring anything about diesel engines. Meanwhile the strong signals are untouched: Age moves -1,908 → -1,804 → -1,875 € per standard deviation across the three fits. Shrinkage is acting on collinear noise, not on structure.

Stepwise selection was run as a contrast, not a recommendation, using HC3 p-values because Part C rejects constant variance. **The two procedures do not agree:** forward stops at 18 predictors, backward at 21, the sets are not nested, and counting LASSO as a third method only **15 of 45 predictors are kept by all three.** Fuel_Type_Diesel is the clearest case — forward adds it sixth at $p = 1.3 \times 10^{-8}$, backward deletes it second at $p = 0.943$, LASSO zeroes it. On cross-validated error the four sets span 23 €, well inside the noise, and three of them were chosen on the very folds that score them. So no stepwise set becomes the working specification; what the exercise establishes is that roughly half the predictors hold no independent information, and which half you discard depends on the procedure. Regularisation was later re-run on the design the diagnosis produced (Section 4.3), because a penalty belongs on the matrix the model will actually use.

4. Part C — Diagnostics and Remedial Action

Part B produced three models that predict equally well. Part C asks whether the model that produces those numbers is *entitled* to them. Every specification passed through one engine — fit, score out of sample on the fixed folds, test six assumptions, log the result — so any difference between two rows of the log is the specification and never the procedure. Each rule pairs a test with an effect size, because at $n = 857$ the formal tests reject departures far too small to act on: RESET must show both $p < 0.05$ and $F \geq 10$ before linearity counts as failed.

4.1 The baseline diagnosis

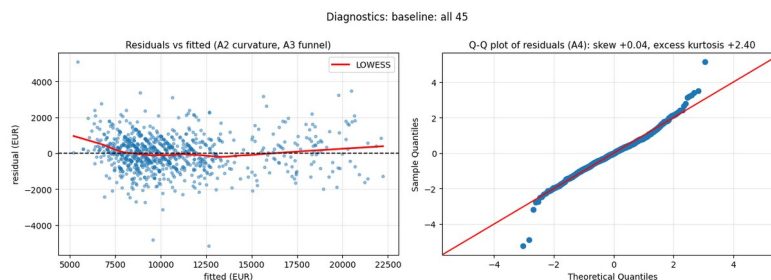


Figure 2. Baseline model, 45 predictors. Left: residuals against fitted values — the LOWESS curve bends and the spread widens with price. Right: Q-Q plot — the residuals are symmetric but heavy-tailed.

Finding — five of six assumptions do not hold. Independence passes (DW = 1.846) and means little, since rows are ordered by identifier rather than by time. Linearity fails at $F = 40.5$ ($p = 1.7 \times 10^{-17}$); homoscedasticity at LM = 234.2; normality at $p = 9.6 \times 10^{-10}$; multicollinearity at max VIF 23.5. Influence is marginal — 7.4% of cars above the $4/n$ screening threshold, but a largest Cook's distance of only 0.144, far below the value of 1 at which one observation would dominate. Prediction is unaffected by any of it: RMSE CV = 1,067.3 €, $R^2 = 0.918$. **So what: the point estimates are usable, the inference around them is not.** That distinction is the whole of Part C. Two details shaped every remedy that followed. Normality fails purely through the *tails* — skew +0.04 against excess kurtosis +2.40 — which points at a robust loss rather than a target transformation. And the multicollinearity has a name: Fuel_Type_Diesel, explained by the other 44 predictors at $R^2 = 0.957$. Ten features sit above VIF 5, but this is one cluster and not ten problems, because diesel Corollas *are* the heavy, large-engine, high-tax cars.

The misspecification is between the variables, not inside them. Partial-residual plots put the largest departure from a straight line at 452 € for KM, 411 € for Weight and 291 € for Age — 3% to 4% of the mean price — while RESET rejects at $F = 40.5$. This contradicts the midterm EDA, which found the price-mileage relationship curved and recommended transforming KM, and both readings are correct: old cars *are* the high-mileage cars, so a two-variable scatter attributes age's curvature to mileage. The recommendation was tested rather than assumed: replacing KM by $\sqrt{\text{KM}}$ costs 13 €, by $\log(\text{KM})$ 68 €, and influence rises to 0.300 because five cars recorded at KM = 1 are sent to zero. **Now what:** curvature living between variables is what an interaction term reaches and a transformation of one column cannot.

4.2 Three remedies, each aimed at a named failure

Remedy 1 — the specification. Squares and products were built beside the original predictors, never in place of them, and on *centred* variables, which keeps the main effect interpretable at the sample mean and removes the x -versus- x^2 collinearity that is an artefact of units. Ten candidates were scored individually, then chosen greedily. Summed they promise 171 €; adding all ten delivers 85 €; forward selection reaches **91 €** with four fewer terms, and the ranking shifts mid-selection — $\log(\text{KM})$ is worth 2.7 € alone but 12.1 € where it enters. The result is the working specification: the 45 base predictors plus $\log(\text{CC})$, $\text{Age} \times \text{Weight}$, $\text{Age} \times \text{KM}$, $\log(\text{KM})$, $\log(\text{HP})$ and HP^2 .

Two controls were run against that conclusion. **Seventeen pure powers make linearity worse** — RESET climbs from 40.5 to 75.2 and prediction deteriorates by 37 € — while six terms including three cross-products cut RESET to 11.3 with eleven fewer columns. And VIF-pruning, the one procedure here that makes multicollinearity pass, **costs 375 €**. Its removal order explains why: *Fuel_Type_Diesel*, then *Weight*, then $\text{Age} \times \text{Weight}$, then *Age* itself — the rule deletes two of the strongest predictors in the data because each is now collinear with its own square and product. **The lesson generalises: VIF pruning must never be run on a design containing a variable together with its own powers and products**, because that collinearity is structural and a VIF rule cannot tell the two cases apart.

Remedy 2 — the response. The textbook treatment for non-constant variance is $\log(\text{Price})$. Predictions were returned to euros using **Duan's smearing factor**, estimated inside each fold, because the naive exponential estimates the conditional median and under-predicts the mean: without the correction the log model is biased by +44 €, with it by -6 €. **Finding: on the assumption it was meant to fix, the log buys almost nothing** — Breusch-Pagan moves from 234.2 to 217.6, a 7% reduction on a statistic that must fall below roughly 60 to stop rejecting. What it does fix is linearity, dropping RESET to 4.67. That is not an argument for adopting it: the interaction terms achieved the same repair while also buying 91 € of accuracy, and stacked on top of them the log leaves RESET unchanged, makes heteroscedasticity worse (136 → 173) and costs 66 €. It also makes normality worse, turning symmetric heavy-tailed residuals into left-skewed heavier-tailed ones (skew -0.38, kurtosis 3.24) — it was aimed at a right-skew the residuals never had.

What survives instead is HC3. Under heteroscedasticity OLS stays unbiased and consistent; what breaks is the variance formula, and with it every standard error and p-value. HC3 uses each observation's own squared residual, weighted by leverage, and changes no coefficient and no prediction. On the working specification it inflates standard errors by a median of **1.18×** and a maximum of **2.43×** on $\log_KM$, and moves **four of 52 coefficients out of significance**. A reader trusting the default output would have drawn four conclusions the data does not support, with no way to know.

Remedy 3 — the loss function. Huber uses squared loss for small residuals and linear loss beyond a threshold, so it targets heavy tails and influential cars. There was a specific prediction to test: if the heavy tails were misspecification rather than genuine outliers, Huber should help the broken model and add nothing to the repaired one. **That is exactly what happened.** On the baseline Huber ties with OLS (1,065.5 against 1,067.3) while down-weighting 151 of 857 cars to a mean weight of 0.73; on the repaired specification it is 7 € worse. Excess kurtosis *rises* under both robust fits, which is the estimator working rather than failing — OLS drags its plane toward a badly mispriced car and artificially shrinks that residual, whereas Huber refuses to chase it.

Quantile regression was run at $\tau = 0.1, 0.5$ and 0.9 for a different purpose: not accuracy, but a question OLS cannot pose. τ is not a confidence level — it estimates a conditional percentile of price, so the three fits describe the cheap, typical and best-kept examples of the same specification. **Finding: only the premium end prices age differently.** The age coefficient is -107.1 €/month at $\tau = 0.1$ and -101.6 at the median — the same number twice against standard errors near 4.5 — but **-120.1 at $\tau = 0.9$** , about 2.7 standard errors steeper; mileage and weight differ well inside the noise, so no claim is made about them. **So what:** for nine cars in ten the single conditional-mean coefficient of -103.6 €/month is the right summary, but at the top of the market it understates monthly depreciation by roughly 16%. Raising the solver's iteration cap tenfold moved no coefficient anywhere, so the estimates are converged.

Specification	k	RMSE CV (€)	RESET F	BP LM	Ex. kurt.	Max VIF
Baseline, all 45	45	1,067.3 ± 20.3	40.5 ✗	234.2 ✗	2.40 ✗	23.5 ✗
+ 3 squares + 3 interactions	51	1,026.8 ± 19.2	11.3 ✗	185.0 ✗	1.23 ✗	27.5 ✗
VIF-pruned to < 5	44	1,402.1 ± 27.9	48.9 ✗	158.7 ✗	1.88 ✗	4.31 ✓
Pure powers (45 + 17)	62	1,104.2 ± 18.7	75.2 ✗	189.4 ✗	1.63 ✗	241.0 ✗
All 10 candidates (45 + 10)	55	982.2 ± 15.8	2.47 ✓	142.9 ✗	0.62 ~	267,969 ✗
Forward-selected 6 (45 + 6)	51	976.2 ± 15.3	16.3 ✗	136.0 ✗	0.60 ~	16,936 ✗
$\log(\text{Price})$, all 45	45	1,064.9 ± 21.7	4.67 ~	217.6 ✗	3.24 ✗	23.5 ✗
Huber (45 + 6)	51	983.1 ± 16.5	0.66 ✓	—	—	16,936 ✗

Table 2. Eight of the nineteen logged runs (full table in Appendix B). ✓ pass, ~ marginal, ✗ fail, — not graded. Robust and quantile fits are not scored on homoscedasticity or normality: they were chosen because those conditions do not hold, so recording them as failures would be a category error.

4.3 What the whole diagnosis says

No specification passes all six. Homoscedasticity fails in every one of the nineteen runs — interactions moved the statistic to 136, a log response to 218, and a robust loss the wrong way to 155. That is not a failure of the search; it is the finding, and the answer is HC3. Multicollinearity passes exactly three times **and every time by deletion**, at a cost of 426 €, 282 € and 68 €; there is no run here that is both clean on collinearity and competitive on error. Linearity is the one failure that genuinely responds to treatment, and normality turned out to be linearity in costume — excess kurtosis tracks the specification almost exactly, from 2.40 at the baseline to 0.60 once the interaction terms are in.

The central tension, stated plainly. The best-predicting specification is also the worst on collinearity: 976 € with max VIF 16,936, against 1,402 € for the only design that makes VIF pass. The study cannot have both, and the rule fixed in advance decides on out-of-sample error rather than on assumption verdicts — the right rule for a prediction problem, recorded here as a limitation rather than hidden. Regularisation was then re-run on that 51-column design, since a penalty treats the damage collinearity does without deleting its cause. **Both grids bottomed out on the boundary at $\lambda = 0.10$** — cross-validation asking for no penalty at all — and LASSO kept all 51 columns; on error the three are the same model (976.2, 978.1, 980.3 € against SE 15). The coefficients tell a different story. Under OLS the design contains pairs that cancel at enormous magnitudes: CC at +11,073 € per SD against log_CC at -10,317, HP at +8,369 against log_HP at -8,057. Ridge collapses HP to 388 and log_HP to -567, cutting total absolute coefficient mass from 48,418 € to 20,979 €. **So what: regularisation here does not repair the diagnostic — VIF is unchanged, because a penalty does not change the columns — it repairs the damage. What it buys is not accuracy but the ability to quote a number.**

5. Model Selection, the Holdout, and What the Model Is Worth

Seventeen candidates were scored under one protocol. Paired Wilcoxon tests on per-car absolute errors — paired on the car and not the fold, because thirty repeated folds reuse the same 857 observations — found **the top of the table statistically indistinguishable** after Holm correction. The correction matters: backward elimination has an uncorrected p of 0.033 and the log model 0.038 — both would have been called significant in isolation, and both become 0.267 once sixteen tests are accounted for. Six candidates do differ, all at the bottom: plain OLS on the base 45 (+43 €), pure powers (+65 €), median regression (+88 €), the compact four-predictor model (+207 €) and the VIF-pruned design (+298 €).

With the data unable to separate the leaders, the pre-registered **one-standard-error rule** decided. The ceiling of 991.5 € held five candidates. Four estimate 52 parameters and one estimates 56, which eliminates the ten-term design at the first criterion even though it is the only specification that satisfies linearity outright. None uses a log target; Ridge and LASSO fall at the third criterion, leaving the forward-selected six terms and Huber tied, and lowest CV error breaks it. **The preferred model was named here**, the candidate list hashed and closed, and only then was the holdout opened.

Candidate	Params	RMSE CV (€)	RMSE hold. (€)	Optimism	MAE hold.	R ² hold.
Forward-selected 6 (45+6) — preferred	52	976.2	1,088.6	+112.4	819.7	0.900
LASSO on 45+6	52	978.1	1,088.8	+110.7	818.6	0.900
Ridge on 45+6	52	980.3	1,086.4	+106.1	816.8	0.900
All 10 candidates (45+10)	56	982.2	1,082.1	+99.9	813.1	0.900
Huber (45+6)	52	983.1	1,086.5	+103.4	815.1	0.900
OLS, all 45 (baseline)	46	1,067.3	1,123.2	+55.9	847.0	0.900

Table 3. Cross-validated and holdout performance, 572 held-out cars scored once. The full seventeen-candidate table is in Appendix C.

Finding — the preferred model comes fourth, by 6 €. Five models sit within 7 € of each other on 572 cars. Under the pre-registered rule that is not a re-crowning: the paired tests had already called them indistinguishable, and picking the holdout winner after the fact would convert 572 protected cars into a second validation set. The rank correlation between CV and holdout is **0.850**, so the protocol ranked the field well even where it could not separate the top of it.

The optimism is measured rather than asserted, and it is the price of selection. The five specifications whose terms were chosen on the folds lose 100–112 € between CV and holdout; models that selected nothing lose far less — plain OLS 56 €, Ridge 54 €, log(Price) on the base design only 26 €. One blemish is systematic and is reported rather than explained away: **every candidate under-prices the holdout**, with mean error from +81 € to +109 € against near-zero bias in cross-validation. It appears under every estimator and specification, so it is a property of the split rather than of any model.

5.1 Practical reading for a dealership

- Operating tolerance. The model prices a Corolla to within roughly 8% on average (MAE 820 €). That is the band a listing decision should be made in; a point estimate carries no more precision than that.
- Correct for the known bias. On unseen cars the model runs about 100 € low, so that offset should be added rather than the raw prediction treated as unbiased.
- Age dominates, and it is not one number. Each month costs roughly 101 € across most of the market but 120 € at the premium end, so holding a well-kept, higher-priced car costs more than the average model implies.
- Data quality beats model choice. The gap between the best and worst estimator here is smaller than the 102 € that two mistyped weight records cost — auditing incoming stock records is a cheaper improvement than a better algorithm. And if a coefficient must be quoted rather than a prediction used, Ridge on the same specification costs 4 € of accuracy and returns readable numbers where OLS returns cancelling pairs of $\pm 10,000$ €.

6. Part D — From Prediction to Causal Inference

Suppose the goal were not prediction but the effect of a single feature on price. The feature chosen is *Automatic_airco* — digital climate control, specified at the factory. Every car has two potential prices, $Y(1)$ with the option and $Y(0)$ without, and only one is ever observed. The target is the average over cars that actually have it, and the observed gap decomposes exactly into that quantity plus a **selection bias** term: what the treated cars would have been worth without the option, compared with cars that never had it. Random assignment is what makes that term vanish, and nothing randomised this option — it was specified in 2002 on cars the manufacturer already intended for the upper end of the range. One clarification is worth stating explicitly: the 60/40 split was random, which makes the two evaluation sets comparable to each other, but says nothing about whether cars with the option are comparable to cars without it. The split was randomised; the treatment was not. The analysis below uses all 1,429 cars, since no model is being selected here and the holdout was already opened. **76 cars (5.3%) carry the option**, at a median price of 19,225 € against 9,750 €.

Specification	k	Coefficient (€)	HC3 SE	p
Raw gap, no controls	1	8,622	396	< 0.001
OLS, all 45 base predictors	45	1,826	284	< 0.001
Forward selection (18)	18	1,975	242	< 0.001
LASSO (45), shrunk	37	1,881	—	—
OLS, 45 + 6 — the preferred model	51	1,210	266	< 0.001

Table 4. What *Automatic_airco* is worth according to five defensible specifications, full sample, HC3 standard errors.

Finding — the controls absorb four-fifths of the gap, and what is left depends on which controls you pick. Adding the 45 base predictors removes 6,796 € of the 8,622 €, or 79%. The remainder is not stable: it ranges from 1,210 € to 1,975 € across four specifications, a spread of 765 € — roughly three times either standard error. Those standard errors describe sampling variability *at a fixed specification*; they say nothing about variation across specifications, which is why the intervals do not overlap in the way their widths suggest. All four were chosen on predictive grounds.

6.1 Three checks on whether any of those numbers is an effect

Reading a coefficient causally requires conditional ignorability and common support. The second is checkable, so it is examined first — if it fails, the first never gets its turn. **Positivity survives on paper and fails in practice.** No trim level is empty, so the condition is not formally violated, but SOL is 10% of the sample and carries **55 of the 76 treated cars**. Outside SOL the entire treated group is 21 cars: 15 unlabelled, 4 LUNA, and **2 TERRA out of 896**. The treatment rate runs from 0.2% to 38.5% across four levels of one covariate. So the estimate is not built from 1,429 comparisons; it is a comparison inside SOL — 55 treated against 88 untreated — plus an extrapolation across 1,286 cars resting on 21 observations. In a region that thin the answer is largely determined by functional form, and functional form is exactly what Part C chose on predictive grounds.

Covariate	Role	Treated mean	Untreated mean	SMD
Age_08_04 (months)	confounder	24.7	57.9	-2.09
Trim_SOL	confounder	0.72	0.07	+1.82
Weight (kg)	confounder	1,148.6	1,066.2	+1.73
Airco	bundled option	1.00	0.48	+1.48
KM	confounder	30,418	70,913	-1.36
Boardcomputer	bundled option	0.82	0.27	+1.32

Table 5. The six largest standardised mean differences. Twelve of sixteen covariates exceed $|SMD| > 0.25$. Full table in Appendix E.

The two groups are not comparable on anything. Age is the extreme case: treated cars average 24.7 months against 57.9, a gap of nearly three years and more than two pooled standard deviations. A regression does not compare like with like — it fits a surface and reads a coefficient off it, and returns a number regardless of whether comparable cars exist. With an SMD of -2.09 on age, the treatment coefficient absorbs whatever a *linear* age term fails to capture over a 33-month displacement, and Part C already showed the age–price relation is not linear.

The bias decomposes exactly, and half of it is age. For OLS the algebra is exact: the 6,797 € removed by the controls is reproduced to the euro by the sum over covariates of β times the treated–untreated difference. **Age alone contributes 3,342 €, 49% of the whole bias** — a treated car is 33.2 months younger and each month is worth 100.72 €. Weight adds 1,585 €, mileage 628 €, horsepower 539 €; those four account for 90%. CC contributes -332 €, the opposite sign, so along that dimension the raw gap understates the option. Adding the trim dummies moved the coefficient by only 42 € while raising the standard error from 241 € to 284 €; trim was retained as a pre-treatment confounder at that modest, reportable cost.

Two further sources of bias cannot be tested with this file at all. **Bad controls:** Airco, Boardcomputer, CD_Player and Powered_Windows are not causes of digital climate control but fellow members of the same factory package. Conditioning on them compares a SOL car with the package to a SOL car with most of it, stripping away part of what the buyer is paying for; any adjustment set is therefore a choice between two errors. **Sample selection and measurement:** every row is a car *offered for sale* in 2004, not a car sold. The listing price is not the transaction price, and a car that sold quickly never entered the file — so if the option shortens time on the lot, the data disproportionately retains the equipped cars that did *not* sell, making selection depend on both treatment and outcome. No diagnostic here would reveal it.

6.2 What data would be needed

The checks above did not find a model that needs fixing; they found a question this file cannot answer. More columns from the same file would not help, because the missing information is not another characteristic of the cars — it is **how the option came to be on them**. Three layers would be collected, each tied to a specific identification strategy.

- The assignment mechanism — the factory build sheet and the option's list price. Today Automatic_airco is a fact about the car with no recorded origin; a build sheet distinguishes a car specified with climate control from one on which it was standard, two different treatments now recorded as the same 1. The list price also gives a benchmark: an estimate far above it would signal the coefficient still carries the trim package.
- Missing confounders and a correctly measured outcome — transaction price and days on the lot dissolve both measurement problems above; condition and service history sit inside the residual; and regional demand is a common cause of both option and price, since a dealer stocks equipment for the customers who also set the local price level.
- Exogenous variation, which changes the design rather than the covariate list — a model year in which the package moved from optional to standard supports difference-in-differences; dealer-level availability driven by supply is a candidate instrument; a tax or trim boundary above which the option was standard supports regression discontinuity.

None of these returns the effect for every Corolla, so the estimand is a **local average treatment effect** — and that locality is already visible, since the comparison lives inside SOL. A well-identified statement about a defined subgroup is worth more than a precise-looking coefficient about all 1,429 cars that no assumption supports.

7. Limitations

- No coefficient of the preferred model reads on its own: CC and log_CC sit at $+11,073$ and $-10,317$ € per standard deviation, cancelling. Only the fitted surface is identified, not its parts.
- Homoscedasticity fails, so classical standard errors are wrong; every statement of significance here uses HC3, which inflates them by $1.18\times$ at the median and up to $2.43\times$.
- One coefficient, one market: the age penalty is 120 €/month at the premium end against 102 € at the median, and a conditional-mean model reports a single number.
- The choice among the top five rests on a rule fixed in advance rather than evidence of superiority, since neither CV nor the holdout could separate them. Two categorical levels are thin — CNG (9 training cars) and Color_Other (4) — so no conclusion rests on them.

The honest summary. What the diagnosis bought is not a better model. It is knowing precisely what this model can and cannot be asked — and Part D is the sharpest form of that: the prediction model answers its own question well, and no rearrangement of these 45 columns answers the causal one.

Appendices

A — Full Part A decision log. B — All nineteen runs with every assumption statistic. C — Seventeen-candidate registry, CV and holdout. D — Sensitivity checks. E — Full balance table and OVB decomposition. F — HC3 versus classical standard errors. G — Executable notebook.